

# Group Theory Notes

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## Group Theory Notes

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Introduction

Examples of Groups

Permutations

Rubik's cube group

2026-06-26

└ Outline

Introduction

Examples of Groups

Permutations

Rubik's cube group

2026-06-26

Group Theory Notes  
└ Introduction

Introduction

# Introduction

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## Example (Rubik's cube)

Consider the Rubik's cube (for an online example, see [here](#)). What moves can we perform on a Rubik's cube?

## Rules of moves

There are many, *many* different configurations (or states) possible, but only one solution. We can get from one state to another by performing a valid movement of the cube. These movements have the following properties:

1. There is an unchanging list of allowable moves.
2. Every move is reversible.
3. Every move is deterministic.
4. Moves can be combined.

2026-06-26

## Group Theory Notes

### └ Introduction

### └ Rubik's Cube

#### Example (Rubik's cube)

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### Example (Generators of Rubik's cube motions)

The website above uses twelve basic moves:  $F$ ,  $F'$ ,  $U$ ,  $U'$ , and so on.

- How can you write  $F'$  in terms of  $F$ ?
- What is the smallest number of moves you can find that represent all of the basic moves on a Rubik's cube?

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- How can you write  $F'$  in terms of  $F$ ?
- What is the smallest number of moves you can find that represent all of the basic moves on a Rubik's cube?

The observations above form the basis of a mathematical structure called a **group**. Groups appear throughout mathematics and its applications.

2026-06-26

## Group Theory Notes

└ Introduction

└ Groups

Groups

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2026-06-26

Group Theory Notes  
└ Examples of Groups

Examples of Groups

## Examples of Groups

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**Example (Coins in a line)**

Consider placing a penny and a nickel side by side in front of you. Is the action of swapping the positions of both coins a group? Note that you will need to specify an appropriate list of moves and determine if that list satisfies the above rules.

**Example (Marbles in piles)**

Suppose you have five marbles in each of two piles in front of you. You have two actions available to you: you can take a marble from the left pile and move it to the right pile, *or* you can take a marble from the right pile and add it to the left pile. Is this a group?

Swaps

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## Example (Three letter words)

Consider all possible three letter "words" that can be formed from the letters  $\{A, B, C\}$ , where each letter must be used precisely once.

1. How many words are possible?
2. Consider two actions: swapping the first and second letter of a word, and swapping the second and third letter of a word. Can these two actions alone generate all of the words possible starting from  $ABC$ ?
3. Is this a group?

2026-06-26

## Group Theory Notes

### └ Examples of Groups

### └ Actions on Words

#### Example (Three letter words)

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End of class <2026-06-22 Mon>

At this point, you might start thinking of a group as a set of objects that we can move around in a nice way. Group theory is about the study of such sets following the rules outlined above.

## Example (Coins in a line (again))

Consider the group of coins in a line.

1. By rule 4, the action from this exercise when performed twice in a row yields another action in the group. What is this action?
2. Does every group have an action like this?

2026-06-26

## Group Theory Notes

### └ Examples of Groups

### └ Group Theory

As a reminder, the group rules are as follows:

1. There is an unchanging list of moves.
2. Every move is reversible.
3. Every move is deterministic.
4. Moves can be combined.

At this point, you might start thinking of a group as a set of objects that we can move around in a nice way. Group theory is about the study of such sets following the rules outlined above.

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1. By rule 4, the action from this exercise when performed twice in a row yields another action in the group. What is this action?
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## Example (Symmetry groups)

Describe the symmetry group of an equilateral triangle and a non-square rectangle.

2026-06-26

Group Theory Notes  
└ Examples of Groups

└ Examples of Groups

Examples of Groups

Example (Symmetry groups)

Describe the symmetry group of an equilateral triangle and a non-square rectangle.

Recall that the symmetry group of a shape is the set of reflections/rotations that leaves the shape unchanged.

## Example (Groups with restrictions)

Try to find or construct groups that satisfy the following constraints:

1. The order in which the actions are performed matters.
2. The order in which the actions are performed doesn't matter.
3. There are exactly three actions.
4. There are exactly four actions.
5. There are infinitely many actions.

2026-06-26

## Group Theory Notes

### └ Examples of Groups

### └ Groups with Certain Properties

1. Rubik's cube group or symmetry group of triangle/square
2. Symmetry group of non-square rectangle or coin group
3. Look at triangle
4. Look at square or non-square rectangle
5. Can use the integers or an infinite chain

#### Example (Groups with restrictions)

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# Nonexamples of Groups

## Example (Sets that break group rules)

Devise a situation that satisfies all of the group rules except for the second, third or fourth depending on your group.

2026-06-26

## Group Theory Notes

### └ Examples of Groups

### └ Nonexamples of Groups

#### Example (Sets that break group rules)

Devise a situation that satisfies all of the group rules except for the second, third or fourth depending on your group.

Break class into three groups (count off 1, 2, 3) and have the groups focus on breaking a single rule from group rules.

1. Breaks second rule: can use naturals with addition or an action that "erases" information.
2. Think of symmetry group where we rotate or flip based on results of coin toss.
3. We already have an example of this. We can also look at limited symmetry groups (i.e., subgroups that aren't closed).

2026-06-26

Group Theory Notes  
└ Permutations

Permutations

# Permutations

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- A **permutation** of an ordered collection of objects is a rearrangement of those objects.
- Every group action we've looked at so far is an example of a permutation.

## Example (Permutations and groups)

List the permutations and the objects being permuted for the following groups:

1. The three letter word group (see Example (Three letter words)).
2. The symmetry group of the equilateral triangle (see Symmetry Groups (slide)).

2026-06-26

## Group Theory Notes

### └ Permutations

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List the permutations and the objects being permuted for the following groups:

1. The three letter word group (see Example (Three letter words)).
2. The symmetry group of the equilateral triangle (see Symmetry Groups (slide)).

- Shapes have symmetry groups (also called *dihedral groups*).
- Collections of objects also have symmetry groups. The actions are permutations.

## Example (Elements of a symmetry group)

Give two elements of the symmetry group of  $\{1, 2, 3, 4\}$ .

## Example (Number of elements of a symmetry group)

How many elements are in the symmetry group above?

2026-06-26

## Group Theory Notes

### └ Permutations

### └ Symmetry Groups

Note the connection to previous permutations!

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Give two elements of the symmetry group of  $\{1, 2, 3, 4\}$ .

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How many elements are in the symmetry group above?



- Two permutations on the same set of elements can be combined.

## Example (Combining permutations)

Let

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 4 & 6 & 2 & 1 \end{pmatrix}, \rho = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 6 & 2 & 5 \end{pmatrix}$$

Find  $\sigma\rho$ .

2026-06-26

Group Theory Notes

└ Permutations

└ Composition of Permutations

Composition of Permutations

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### Example (Combining permutations)

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Find  $\sigma\rho$ .

- The permutation  $\sigma \in S_3$  given by

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

can be written as (231).

## More cycles

Write the permutations  $\sigma$  and  $\rho$  from Example: Combining permutations using cycle notation.

2026-06-26

## Group Theory Notes

### └ Permutations

### └ Cycle Notation

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#### More cycles

Write the permutations  $\sigma$  and  $\rho$  from Example: Combining permutations using cycle notation.

- The symmetry group on  $n$  elements is called  $S_n$ .
- You've already found  $S_3$ !
- Permutations in  $S_n$  can be written in cycle notation.

## Example (Cycle notation in $S_4$ )

Write the permutations you found in Example (Elements of a symmetry group) using cycle notation.

2026-06-26

Group Theory Notes

└ Permutations

└ Cycle Notation Again

Cycle Notation Again

Example (Cycle notation in  $S_4$ )

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# Combining Permutations Again

- Cycle notation provides a convenient way to combine permutations.

## Example (Combining permutations with cycles)

Find  $\sigma\rho$  using cycle notation.

2026-06-26

Group Theory Notes

└ Permutations

└ Combining Permutations Again

Combining Permutations Again

- Cycle notation provides a convenient way to combine permutations.

Example (Combining permutations with cycles)

Find  $\sigma\rho$  using cycle notation.

A *Cayley table* is a multiplication table for a group.

**Example (Cayley table for symmetry group)**

Find the Cayley table for the symmetry group of the rectangle

Example (Cayley table for  $S_3$ )

Find the Cayley table for  $S_3$  (i.e., the symmetry group of the equilateral triangle).

# Even and Odd Permutations

- Permutations can be categorized as **even** or **odd** based on how many switches they perform
- This is called the **parity** of the permutation

## Example (Determine parity in $S_3$ )

Find the parity of the permutations (12) and (123).

## Example (Determine parity in $S_6$ )

Find the parity of the permutations  $\sigma$ ,  $\rho$  and  $\sigma\rho$ .

2026-06-26

## Group Theory Notes

### └ Permutations

### └ Even and Odd Permutations

Even and Odd Permutations

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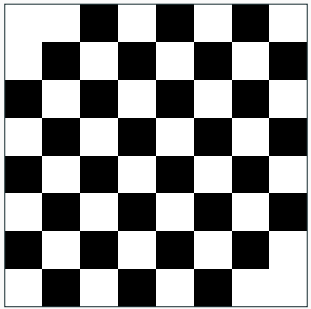
- Many classic puzzles (including the Rubik's Cube) can be described in terms of permutations
- Knowing something about the allowed permutations tells us about the puzzle itself!
- These puzzles often have **invariants** that we can use to analyze them

- The Mutilated Chessboard Puzzle asks if it is possible to cover a chessboard with two opposite corners removed using  $2 \times 1$  dominoes.

Figure 1: Chess board with opposite diagonal corner pieces missing.



**Figure 1:** Chess board with opposite diagonal corner pieces missing.



# The 15-Puzzle

- The goal is to solve the puzzle to put the numbers in order (i.e., swap the 14 and 15 squares)
- The blank square must be on the bottom right at the end

|    |    |    |    |
|----|----|----|----|
| 1  | 2  | 3  | 4  |
| 5  | 6  | 7  | 8  |
| 9  | 10 | 11 | 12 |
| 13 | 15 | 14 |    |

**Figure 2:** The 15-puzzle

2026-06-26

## Group Theory Notes

- └ Permutations

- └ The 15-Puzzle

The 15-Puzzle

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Figure 2: The 15-puzzle

# "A Pestilence"

*FIFTEEN.*

No pestilence has ever visited this or any other country which has spread with the awful celerity of what is popularly called the "Fifteen Puzzle." It is only a few months ago that it made its appearance in Boston, and it has now spread over the entire country. Nothing arrests it. Neither age nor sex is spared by it, and it now threatens our free institutions, inasmuch as from every town and hamlet there is coming up a cry for a "strong man" who will stamp out this terrible puzzle at any cost of Constitution or freedom.

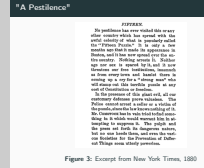
In the presence of this giant evil, all our customary defenses prove valueless. The Police cannot arrest a seller or a victim of the puzzle, since the law knows nothing of it. Mr. Comstock has in vain tried to find something in it which would warrant him in attempting to suppress it. The pulpit and the press set forth its dangerous nature, but no one heeds them, and even the various Societies for the Prevention of Different Things seem utterly powerless.

Figure 3: Excerpt from New York Times, 1880

2026-06-26

Group Theory Notes  
└ Permutations

└ "A Pestilence"



## Example (Model a configuration)

Use a permutation to describe the following arrangement of the 15-puzzle:

|    |    |    |    |
|----|----|----|----|
| 8  | 12 | 2  | 5  |
| 11 | 1  | 6  |    |
| 7  | 14 | 10 | 15 |
| 9  | 4  | 3  | 13 |

**Figure 4:** A shuffled 15-puzzle

2026-06-26

## Group Theory Notes

### └ Permutations

### └ Permutations and the 15-Puzzle

#### Example (Model a configuration)

Use a permutation to describe the following arrangement of the 15-puzzle:

|    |    |    |    |
|----|----|----|----|
| 8  | 12 | 2  | 5  |
| 11 | 1  | 6  |    |
| 7  | 14 | 10 | 15 |
| 9  | 4  | 3  | 13 |

Figure 4: A shuffled 15-puzzle

## Example (Find parity of configurations)

Determine the parities of the permutations corresponding to the puzzles in the previous examples, and find the parity of the solution of the 15-puzzle.

## Parity of moves in the 15-puzzle

Every valid configuration in the 15-puzzle (that is, moves that place the empty spot in position 16) corresponds to an even permutation.

### Example (Find parity of configurations)

Determine the parities of the permutations corresponding to the puzzles in the previous examples, and find the parity of the solution of the 15-puzzle.

### Parity of moves in the 15-puzzle

Every valid configuration in the 15-puzzle (that is, moves that place the empty spot in position 16) corresponds to an even permutation.

## Theorem (Nonexistence of solution)

*There is no solution of the 15-puzzle.*

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*There is no solution of the 15-puzzle.*



# Variation of the 15-Puzzle

## Example (Empty square on bottom left)

Suppose we don't want the empty square to end up on the bottom left, but rather the bottom right:

|   |    |    |    |
|---|----|----|----|
| 1 | 2  | 3  | 4  |
| 5 | 6  | 7  | 8  |
| 9 | 10 | 11 | 12 |
|   | 13 | 14 | 15 |

**Figure 5:** The 15-puzzle

Is the puzzle now solvable?

2026-06-26

## Group Theory Notes

### └ Permutations

### └ Variation of the 15-Puzzle

Variation of the 15-Puzzle

#### Example (Empty square on bottom left)

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| 1 | 2  | 3  | 4  |
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Figure 5: The 15-puzzle

Is the puzzle now solvable?

2026-06-26

Group Theory Notes  
└─ Rubik's cube group

Rubik's cube group

## Rubik's cube group

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The **cubies** on a Rubik's cube can be placed in the following categories:

1. Corner cubies
2. Edge cubies
3. Center cubies

2026-06-26

Group Theory Notes  
└ Rubik's cube group  
└ Terminology

Terminology

The **cubies** on a Rubik's cube can be placed in the following categories:  
1. Corner cubies  
2. Edge cubies  
3. Center cubies

- A **cubicle** is the position of a given cubie. Moves changes the positions of the cubies, but *not* the cubicles.
- Positions can be specified by listing incident faces: `ufr` represents the cubicle/cubie on the upper front right face (i.e., a corner).

## Example (Edge Positions)

Give all positions for edge cubies on the back face of a cube.

2026-06-26

Group Theory Notes  
└ Rubik's cube group

└ Cubies and Cubicles

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Example (Generators Acting on Cubies)

What happens to cubies in the previous categories if you apply any of the six basic Rubik's cube actions?

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# Estimating Rubik's Cube Group

2026-06-26

## Group Theory Notes

### └ Rubik's cube group

### └ Estimating Rubik's Cube Group

#### Example (Size of the Rubik's cube group)

The Rubik's cube group is based on the six fundamental motions F, B, U, D, L and R. How large could this group be? *Hint:* try answering the following questions.

1. How many different positions can the corner cubies be in?
2. How many different positions can the edge cubies be in?

#### Example (Size of the Rubik's cube group)

The Rubik's cube group is based on the six fundamental motions F, B, U, D, L and R. How large could this group be? *Hint:* try answering the following questions.

1. How many different positions can the corner cubies be in?
2. How many different positions can the edge cubies be in?

1. The corner cubies can be arranged in up to  $8!$  ways with up to 3 different configurations each, giving an upper bound of  $8!3^8$  arrangements for the corner cubies.
2. The edge cubies have an upper bound of  $12!2^{12}$  arrangements.
3. The upper bound for moves is therefore  $8!3^812!2^{12}$ .